3D-PRINTED ENCLOSURE GUIDE · SKU CQA240229G16

Pi 5 Case with Headset-Mount Lid

A printable case that holds a Raspberry Pi 5 with PiEEG-16 and mounts on the back of the open-source 3D-printable IronBCI EEG headset, so the whole system can be worn.



2	96.6 × 65	94 × 86	M2.5 × 12	0.12 mm
Printed parts	mm base footprint	mm lid with wings	Lid mount screws	Lid layer height

01 PARTS AND FILES

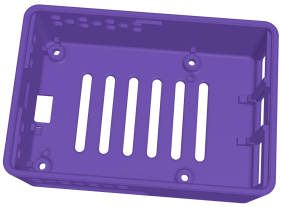
Part	File	Size (mm)	Volume
Case body — Raspberry Pi 5	`RPi 5 case airvents slim ports +6mm.stl`	96.6 × 65.0 × 35.5	34.4 cm³
Headset-mount lid	`pieeg-headset-case-lid-mount.stl`	94.1 × 86.0 × 21.4	11.2 cm³

Files: github.com/pieeg-club/PiEEG-16 → 3D_Print_Headset_Case. Case body remixed from a Printables Raspberry Pi 5 case; headset: open-source IronBCI 3D-printable EEG headset (Instructables).

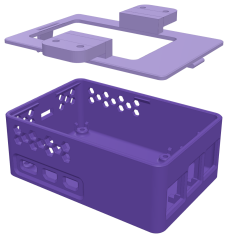
02 VIEWS



Case body — vents, Pi 5 port cut-outs



Case body from above — stand-offs

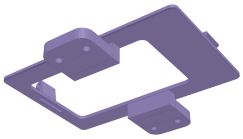


Lid with mounting wings above the body

For research and education only. Not a medical device. Specifications subject to change.



Headset-mount lid — latch and wings



Lid underside — wing screw holes



Pi 5 + PiEEG-16 inside the case

03 PRINT SETTINGS

Setting	Case body	Headset-mount lid
Orientation	Bottom side down	Wing side down (latch side up)
Supports	Little or none	Standard supports
Infill	15–20 %	30 % gyroid
Layer height	0.2 mm	0.12 mm
Material	PLA or PETG (non-conductive)	PETG recommended for the wings; PLA works
Approx. filament	≈ 25–35 g	≈ 8–12 g

Orientation, supports, infill and layer height for the lid are from the design notes; other values are typical starting points.

04 ASSEMBLY

1 Fit the Pi and shield

Attach the Raspberry Pi 5 and PiEEG-16 in the case body with standard stand-offs and mounting hardware. Fit the shield only with the Pi unpowered.

2 Mount the lid to the headset

Fix the lid's wings to the headset with **M2.5 × 12 screws and nuts**.

3 Close the case

Clip the case body onto the lid, routing the electrode leads to the two headers.

4 Secure

If needed for extra stability, add a Velcro strap around the unit.

WARNING Wearable does not mean mains-safe.

On the headset, the Pi and PiEEG-16 must still run from a 5 V power bank only, with every connected device on battery.

05 RELATED DOCUMENTS

PE16-DS-001 Product Datasheet	**PE16-MD-002** Mechanical & Dimensional Drawing	**PE16-PCB-003** PCB & Manufacturing Specification
PE16-PIN-004 Connector Pinout & Electrode Guide	**PE16-QSG-005** Quick Start & Safety Guide	**PE16-SW-006** Data Interface & Software Guide

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